

Zijian Qin

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EDUCATION

Princeton University

- Ph.D. in Computer Science, advised by Prof. Wyatt Lloyd 08/2024 – 05/2029

Nanjing University

- B.Eng. in Electronic Science and Engineering, GPA: 4.59/5.00 (top 1%) 09/2020 – 07/2024

PUBLICATIONS

- Christopher Hodsdon, **Zijian Qin**, Khiem Ngo, Siddhartha Sen, Ethan Katz-Bassett, and Wyatt Lloyd. “Avicenna: Masking Slowdowns in Replicated State Machines with Counterfactual Evaluation.” *In EuroSys 2026*. **Best Paper Award**. [\[paper\]](#) [\[code\]](#)
- Xinle Jia, Yangcan Zhou, **Zijian Qin**, Jingjing Zhu, and Zhongfeng Wang. “Reduced Complexity Dynamic Schedule for Layered Decoding of LDPC Codes.” *IEEE Transactions on Vehicular Technology*, 74(8):13156–13160, 2025. [\[paper\]](#)
- Yangcan Zhou, **Zijian Qin**, and Zhongfeng Wang. “Efficient Soft-Cancellation Flip Decoding of Polar Codes.” *IEEE Communications Letters*, 28(8):1745–1749, 2024. [\[paper\]](#)
- **Zijian Qin**, Yangcan Zhou, Jingjing Zhu, and Zhongfeng Wang. “An Efficient Puncturing Scheme for Polar Codes Based on Fixed Information Set.” *In ICC 2023*. [\[paper\]](#)
- **Zijian Qin**, Yangcan Zhou, and Zhongfeng Wang. “A Second Minimum Approximation Method for Layered Min-Sum Decoding of QC-LDPC Codes,” *In ICTC 2023*. [\[paper\]](#)

EXPERIENCE

Princeton University

Graduate Research Assistant

Princeton, NJ, USA

Aug 2024 – Present

- Advisor: Prof. Wyatt Lloyd.
- Topic: Design, implementation and evaluation of efficient and practical distributed systems.
- **Avicenna**: A replicated state machine protocol that tolerates a slow replica without sacrificing normal-case latency. **[EuroSys '26]** [\[paper\]](#) [\[code\]](#)
 - Designed a counterfactual evaluation mechanism to detect the slow leader.
 - Optimized the leader rotation protocol to reduce the overhead of leader changes.
- **[On going]** A geo-distributed replicated state machine protocol that allows low-latency linearizable local reads.
- **[On going]** An agentic workflow system that makes distributed system evaluation more realistic.

Nanjing University

Undergraduate Research Assistant

Nanjing, Jiangsu, China

April 2022 – July 2024

- Advisor: Prof. Zhongfeng Wang.
- Topic: Code design and decoder optimization for error correction codes in wireless communications.
- Decoder optimization for LDPC codes and polar codes with better error rates and higher efficiency.
- Code design of arbitrary-length polar codes with better error rates.

TEACHING EXPERIENCE

- **Graduate Teaching Assistant, Princeton University**
 - COS 418: Distributed Systems 09/2025 – 12/2025
 - COS 316: Principles of Computer System Design 02/2026 – 5/2026

AWARDS

- **Best Paper Award of EuroSys 2026** 2026
 - 2 / 319 submissions, 2 / 59 accepted papers
- **Chenxue Alumni Scholarship** 2024
 - For academic performance and future studies
- **Xiaomi Scholarship** 2023
 - 1/208 for academic performance
- **First prize of People's Scholarship** 2022
 - Top 5% for academic performance
- **01 Alumni E-scholarship** 2021
 - Top 2% for academic performance